Network access controls — security groups, endpoint policies, and the layers between

Companion to `architecture.md`, answering one question in depth: **who is allowed to talk to what, and which mechanism enforces it.** Three visuals carry the story; the tables give the exact rules. Diagrams are generated by `diagrams/generate.py` — regenerate rather than hand-editing the SVGs.

Accepted risks, up front (SSP-scoped decisions, not oversights):

- The `ecs` interface endpoint carries **no endpoint policy** — GovCloud does not support one there; IAM-side scoping covers it.
- The ALB access-logs bucket is SSE-S3, not CMK — ELB log delivery does not support KMS.

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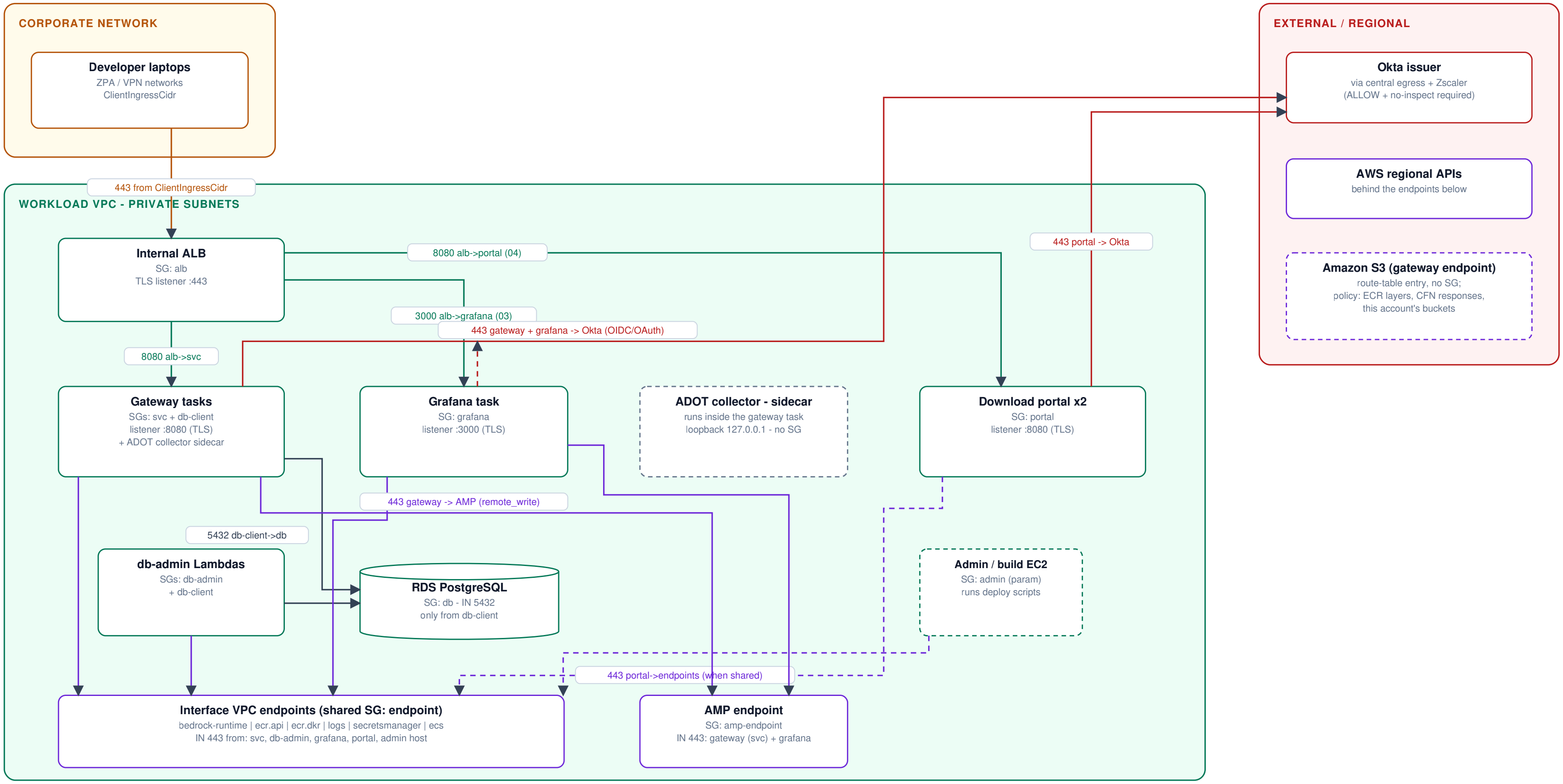
1. The map — who may talk to whom

Every arrow is an explicit SG rule *pair*: an egress rule on the source group **and** an ingress rule on the destination group. A connection with only one half configured does not work; a connection with neither is dropped silently. Every group suppresses default egress with a `127.0.0.1/32` placeholder, so nothing here is reachable "by default."

7. Security groups - who may talk to whom

Every arrow is an explicit SG rule pair (egress on the source + ingress on the destination). Everything else is denied.



Not shown: every SG's default egress is a 127.0.0.1/32 placeholder (deny) unless a rule above exists; endpoint ENIs and RDS initiate nothing. Dashed = optional/conditional (admin-host param, AMP endpoint toggle, portal's shared-endpoint ingress).

Reading tips:

- **Workloads carry more than one SG.** The gateway tasks run with `svc` + `db-client`; the db-admin Lambdas with `db-admin` + `db-client`. Attaching 01's `db-client` group is *the* mechanism for "may reach the database" — nothing else is ever granted 5432.
- **The interface endpoints are the chokepoint.** Their private DNS captures AWS API calls from **every** VPC client, so any in-VPC caller of ecr/logs/secretsmanager/ecs — including the admin/build host — must appear in the endpoint SG's ingress or its calls black-hole (this bit the first test run twice).

- **Okta is the only external dependency** the workloads dial out to; it rides the landing zone's central egress and needs the Zscaler ALLOW + no-inspect prerequisite (see `networking-request-email.md`).

2. The inventory — every rule, by group

8. Security-group rule inventory

Every rule in the deployment, by group. IN = ingress, OUT = egress. Groups not listed here do not exist.

alb (internal ALB) 02

IN 443 from ClientIngressCidr (developer networks)
OUT 8080 to svc (forward + health checks)
OUT 3000 to grafana (rule added by 03)
OUT 8080 to portal (rule added by 04)

svc (gateway tasks) 02

IN 8080 from alb
OUT 443 0.0.0.0/0 (Okta, AWS APIs via endpoints/egress)
OUT proxy port 0.0.0.0/0 (only when HttpsProxyUrl set)
OUT 443 to amp-endpoint (remote_write; rule added by 03)
also carries db-client (5432) + telemetry sidecar (loopback)

db-admin (bootstrap + rotation Lambdas) 02

IN none
OUT 443 0.0.0.0/0 (Secrets Manager, ECS via endpoints)
also carries db-client (below) for 5432

endpoint (shared, all 02 interface endpoints) 02

IN 443 from svc, db-admin
IN 443 from grafana (rule added by 03)
IN 443 from portal (rule added by 04, when shared)
IN 443 from AdminClientSecurityGroupId (optional param)
OUT none (endpoint ENIs never initiate)

portal (download portal task) 04

IN 8080 from alb (path rule /portal, /portal/*)
OUT 443 0.0.0.0/0 (Okta, S3, CloudWatch, ECR, Secrets Mgr)
OUT proxy port 0.0.0.0/0 (only when HttpsProxyUrl set)

db-client (attach to reach the DB) 01

IN none
OUT 5432 to db
attached to: gateway tasks, both db-admin Lambdas

db (RDS instance) 01

IN 5432 from db-client
OUT none

grafana (dashboard task) 03

IN 3000 from alb (path rule /grafana)
OUT 443 0.0.0.0/0 (AMP queries, Okta OAuth, ECR)

amp-endpoint (aps-workspaces endpoint) 03

IN 443 from svc (gateway task - remote_write)
IN 443 from grafana (queries)
OUT none

Cross-stack rule writers

03 adds rules to SGs it imports from 02:
- AlbToGrafanaEgress: alb OUT 3000 -> grafana
- GatewayToAmpEndpointEgress: svc OUT 443 -> amp-endpoint
- GrafanaToEndpointsIngress: endpoint IN 443
04 adds rules to SGs it imports from 02:
- AlbToPortalEgress: alb OUT 8080 -> portal
- PortalToEndpointsIngress: endpoint IN 443 (when shared)
02 param AdminClientSecurityGroupId:
- AdminToEndpointsIngress: endpoint IN 443

Deploy order matters: imported SGs must exist first (02 before 03/04); CREATE_SUPPORTING_ENDPOINTS must match across the deploys.

Reading the map

A connection works only when BOTH ends agree: an egress rule on the source SG and an ingress rule on the destination SG. Every SG here is default-deny; the 127.0.0.1/32 egress entries in the templates are deliberate no-op placeholders that suppress the default allow-all egress.

Resolved (was accepted risk C2): the gateway->collector OTLP hop is no longer on the network - the ADOT collector runs as a co-resident sidecar in the gateway task, reached over loopback (127.0.0.1:4318). No SG rule exists or is needed for it.

SG	Stack	Attached to	Ingress	Egress
`alb`	02	internal ALB	443 from `ClientIngressCidr`	8080→ `svc` ; 3000→ `grafana` (03)
`svc`	02	gateway tasks (incl. the co-resident **ADOT collector sidecar**)	8080 from `alb`	443 anywhere; proxy port (optional); 443→ `amp-endpoint` (03)
`db-client`	01	gateway tasks, db-admin Lambdas	—	5432→ `db`
`db`	01	RDS instance	5432 from `db-client`	—
`db-admin`	02	bootstrap + rotation Lambdas	—	443 anywhere
`endpoint`	02	all 02 interface endpoints	443 from `svc` , `db-admin` , `grafana` (03), admin host (param)	—
`grafana`	03	Grafana task	3000 from `alb`	443 anywhere
`amp-endpoint`	03	aps-workspaces endpoint	443 from `svc` (gateway sidecar), `grafana`	—

There is **no** `collector` **security group**: the ADOT collector is a co-resident sidecar inside the gateway task, reached over loopback (`127.0.0.1:4318`), so its OTLP receiver is never on the network and needs no SG rule (this eliminated the former plaintext gateway→collector hop — security review C2, resolved 2026-07-22). The sidecar's remote-write to AMP rides the gateway `svc` SG.

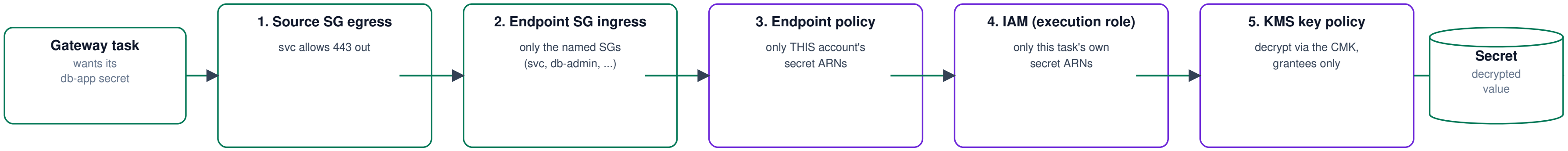
Cross-stack rule writers (03 and the admin-host parameter modify imported 02 SGs as separate `SecurityGroup{In,E}gress` resources): `AlbToGrafanaEgress` , `GrafanaToEndpointsIngress` , `GatewayToAmpEndpointEgress` (the sidecar's remote-write path, which replaced the former `GatewayToCollectorEgress` / `CollectorToEndpointsIngress`), `AdminToEndpointsIngress` . Deploy order (02 before 03) and a matching `CREATE_SUPPORTING_ENDPOINTS` across both deploys are what make these land correctly.

3. The layers — what each control stops that the others can't

SGs are one of **five** gates on the path of a single AWS API call. The diagram traces a gateway task reading its DB secret; the failure table under it is the reason none of the layers is redundant — the second row (attacker-owned credentials exfiltrating *through* a private endpoint) is the one SGs and your own IAM cannot stop alone, and is why every endpoint carries a resource policy.

9. Layered access control - what each layer stops

One request traced through every gate: a gateway task reading its DB secret through the Secrets Manager endpoint.



Why the layers are not redundant - each stops a different failure:

- **Compromised box in another subnet, ANY credentials**
  stopped at layer 2 - its SG is not in the endpoint SG ingress (network path denied before authn)
- **Compromised workload using ATTACKER-OWNED AWS credentials (exfiltration through the private endpoint)**
  stopped at layer 3 - the endpoint policy allows only this account's resources; your IAM cannot judge foreign principals
- **Legitimate workload asking for a secret it does not own**
  stopped at layer 4 - execution roles enumerate exact ARNs (gateway task cannot read the master secret: no task injects it)
- **Principal with a stolen ciphertext or over-broad S3/API access**
  stopped at layer 5 - the CMK key policy grants decrypt only to the roles that need it

Non-SG enforcement points elsewhere in the deployment

bedrock-runtime endpoint policy: pinned to the THREE configured model IDs + their inference profiles (not anthropic.*) | ecs endpoint: NO policy (GovCloud unsupported) - IAM-side scoping covers it

S3 gateway endpoint policy: ECR layer bucket + CloudFormation custom-resource response buckets + this account's buckets only

ALB access-log bucket policy: exactly two writers (ELB log-delivery service principal + legacy regional ELB account) | AMP endpoint policy: this workspace's ARN only

Okta egress: Zscaler policy must ALLOW + not inspect the issuer FQDN for server-originated traffic (org prerequisite - see the networking request)

Endpoint/resource policies in force:

Where	Policy scope
`bedrock-runtime` endpoint	the **three configured model IDs** + their inference profiles — not `anthropic.*`
`ecr.api` / `ecr.dkr` endpoints	auth token (no-resource) + this account's repositories
`logs` endpoint	this account's log groups
`secretsmanager` endpoint	this deployment's secret ARNs + `GetRandomPassword`
`ecs` endpoint	**none** (unsupported in GovCloud) — IAM covers
S3 gateway endpoint	ECR layer bucket, CloudFormation custom-resource response buckets, this account's buckets
`aps-workspaces` endpoint (03)	this AMP workspace's ARN only
ALB access-logs bucket	exactly two writers: ELB log-delivery service principal + legacy regional ELB account

IAM and KMS (layers 4–5) are inventoried in `architecture.md` §6 — the short version: execution roles enumerate exact secret ARNs, the task role holds `bedrock:InvokeModel*` on exactly the three models, and the single CMK's key policy names its grantees.

Change discipline: a new workload that needs the database gets `db-client` attached (never a new 5432 rule); a new in-VPC caller of AWS APIs gets an ingress on the `endpoint` SG (or it will black-hole); anything touching the endpoint SG or the observability SGs should re-check the cross-stack reachability findings in `security-review-2026-07.md`.